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PALAEOMAGNETISM, RADIOMETRIC AGE AND GEOCHEMISTRY OF AN ADAMELLITE AT YETHOLME, N.S.W.

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(With 2 Tables and 3 Figures)

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ABSTRACT

The adamellite that outcrops at Yetholme in central western New South Wales is a satellite body of the Bathurst batholith. Fresh exposures have recently become available in a new road cutting. Abundant potassium-feldspar megacrysts and mafic xenoliths occur in the adamellite and it is cut by aplite and dolerite dykes. The fresh rocks have been sampled for petrological and geochemical studies and for radiometric dating. Palaeomagnetic data from three rock types (adamellite, xenoliths, and dolerite) indicate stability of magnetization. Results of all these studies complement each other.

Radiometric ages (K-Ar) for the adamellite and dolerite confirm a Late Carboniferous age (312 m.y.) of emplacement, as does a Rb-Sr age (325 ± 17 m.y.) determination on the aplite. These confirm three earlier radiometric age determinations for the Bathurst batholith and country rocks and stratigraphic age estimates. The palaeomagnetic directions from these three (Yetholme) rock types agree well with other palaeomagnetic data from Carboniferous rocks from the Newcastle to Tamworth region of New South Wales. The palaeogeomagnetic pole position obtained from the Upper Carboniferous Yetholme adamellite was at 59°S, 150°E ($dp = 5°$, $dm = 8°$).

INTRODUCTION

The western edge of the Sydney Basin overlaps lower, middle and upper Palaeozoic rocks, including the granitic rocks of the Bathurst batholith (Vallance, 1969). This complex batholithic mass outcrops over an area in excess of 1600 km^2. Geological relations (summarized by Vallance, 1969) and age determinations (Evernden & Richards, 1962; Hirt *et al.*, 1963) indicate a Carboniferous age for the batholith.

Considerable spread in palaeomagnetic pole positions occurs between Early and Late Carboniferous rocks of Australia (McElhinny & Embleton, 1974). Additional data on the apparent polar-wander path for the late Palaeozoic could be obtained from the Bathurst batholith if it contained a stable primary remanence (TRM). At Yetholme a new road cutting has been made through an adamellite which is apparently a satellite to the main batholith (Fig. 1). As this road cutting exposed fresh rock it provided an opportunity to investigate the palaeomagnetism of the adamellite, its xenoliths, and cross-cutting aplite and dolerite dykes. The adamellite has a roughly circular outcrop approximately 6 km in diameter, the length of the road cutting being about 0.3 km. Detailed descriptions of the adamellite and the country rock have been given by Mackay (1964) and Watts (1969), and more recently Phillips & Carr (1973) have described aspects of the adamellite's petrology. All four of the rock types in the cutting have been studied petrologically and geochemically and (apart from the xenoliths) have had their radiometric ages determined to complement the palaeomagnetic investigation.

OPAQUE MINERALOGY

In the adamellite, magnetite generally comprises no more than 1%, usually as anhedra up to 0.25 mm in diameter. Although most grains are no more than 0.05 mm in diameter, they commonly occur in aggregates up to 2 mm across. Small magnetite grains (0.005 mm to 0.05 mm) are present in biotite flakes. The magnetite apparently contains little titanium, and oxidation to hematite is slight. In the aplite, the magnetite is very similar to that in the adamellite and in the xenoliths. The magnetite content is greater in the xenoliths, although it varies from xenolith to xenolith. In addition to minor hematite oxidation of the magnetite, rare discrete grains of hematite and maghemite are present in some xenoliths. In the dolerite, titaniferous magnetite and ilmenite comprise 5% to 6% by volume, mainly as small grains from 0.01 mm in diameter down to a size not resolvable by optical microscopy. Grains (and

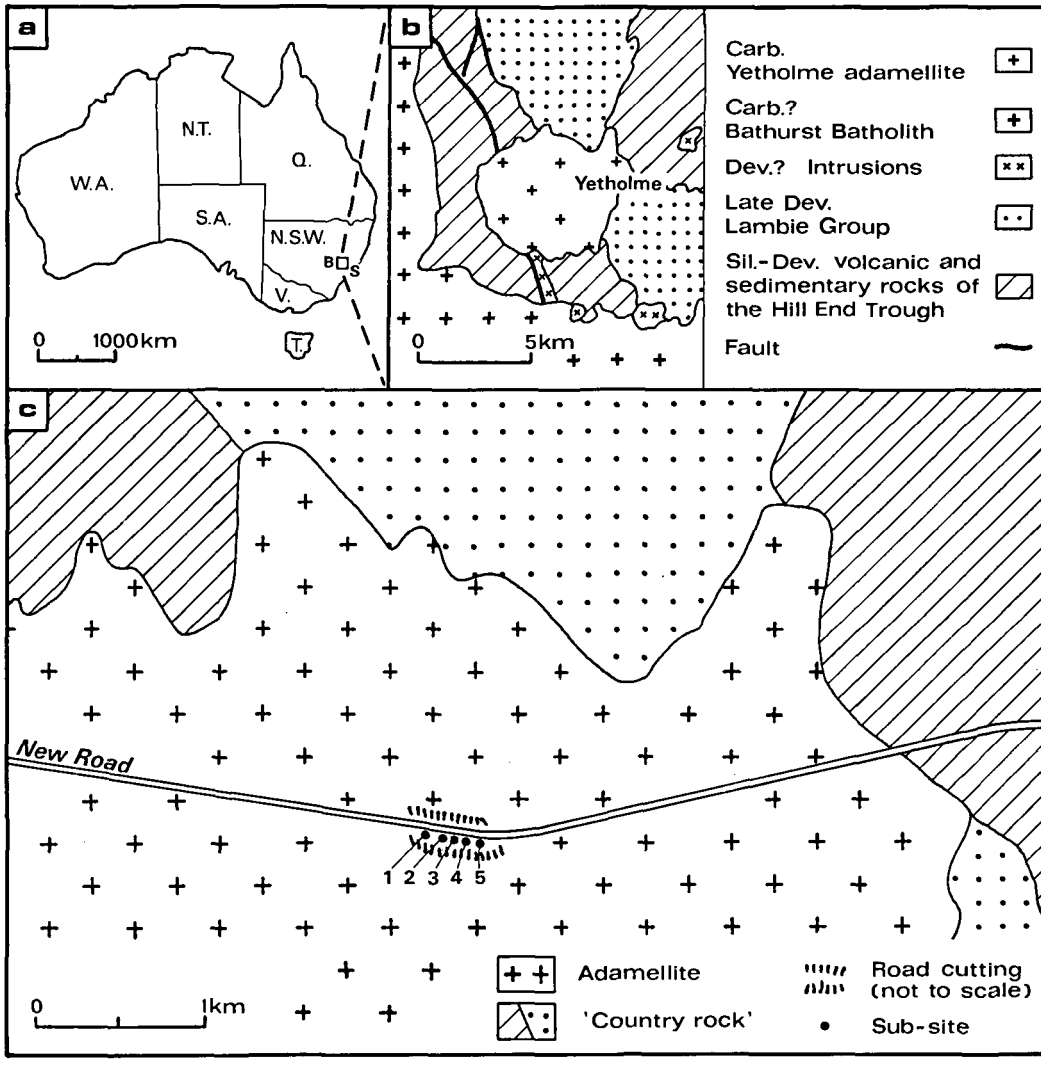


Fig. 1. Locality maps. (a) Geographic location (square) of the sampling locality (S-Sydney, B-Bathurst). (b) General geological setting of the adamellite at Yetholme, modified after Packham (1968). (c) Sampling sub-sites in the Yetholme adamellite (geology based on Mackay, 1964). The mean geographic coordinates of the sampling locality are Lat. 33°27′S, Long. 149°49′E.

needles) 0.02 mm to 0.2 mm in size are also present. Hematite oxidation comprises about 5% of the coarse grains. Sulphite minerals are widespread in the road cutting, but localized along joints and in alteration zones. The sulphides are pyrite and molybdenite of polytype $2H_1$ (I. R. Plimer, *pers. comm.; cf.* Ayres, 1974).

RADIOMETRIC AGES

Evernden & Richards (1962) carried out K-Ar dating on granites from the eastern edge of the Bathurst batholith at Hartley. Hirt *et al.* (1963) used Re-Os techniques to date molybdenite from the skarn just south of the Yetholme adamellite. Table I summarizes those results, which indicate a Carboniferous age for the emplacement of the batholith. Results of radiometric dating carried out for this investigation are also contained in Table I, and these indicate a Late Carboniferous age for the Bathurst batholith. When sampling the dolerite it had been thought that it may have been related to 'Tertiary' basalts which outcrop

TABLE I

Radiometric ages for rocks from (and around) the Bathurst Batholith

Sample Description	Locality	Method	Age, m.y.	Reference
Granite	Hartley	K-Ar	300*	Evernden & Richards (1962, p. 17)
Granite	Hartley	K-Ar	308*	Evernden & Richards (1962, p. 17)
Molybdenite (from a skarn)	Yetholme (?Mt Tennyson)	Re-Os	330 ± 40	Hirt *et al.* (1963, p. 40)
Biotite from adamellite (sub-site 4, 50 m W of dolerite)	Yetholme	K-Ar	309 ± 8	This study
Biotite from adamellite (sub-site 5, 5 m W of dolerite)	Yetholme	K-Ar	312 ± 8	This study
Aplite (whole rock) (Sub-site 3)	Yetholme	Rb-Sr	325 ± 17	This study
Dolerite (whole rock) (Sub-site 5)	Yetholme	K-Ar	298 ± 10	This study

The K-Ar ages for this study were determined by courtesy of Dr D. C. Green (who considered the 312 ± 8 date to be the most reliable—although they are all very similar). The Rb-Sr age was determined by The Australian Mineral Development Laboratories (isochron using 4 samples of aplite and 1 sample of contact adamellite; $^{87}Rb\lambda = 1.39 \times 10^{-11}/y$; $^{87}Sr/^{86}Sr = 0.7051 \pm 0.0016$).

* Note added in proof: Recently Cas *et al.* (1976, p. 205) have recalculated these ages as 310 and 320 m.y. respectively, based on 'preferred decay constants'.

throughout eastern New South Wales. Although some radiogenic Ar loss has probably occurred (with the observed alteration of the dolerite), the calculated age is still Late Carboniferous. Thus, although the dating techniques and the rock types studied vary, all radiometric age determinations indicate a Late Carboniferous age.

PALAEOMAGNETIC RESULTS

Sampling and Preliminary Results

Outcrop suitable for palaeomagnetic study is limited to one road-cutting near Yetholme. Although this cutting (about 0.3 km in length) would often be considered as one extended palaeomagnetic sampling site, following the procedure of Robertson & Watkinson (1974), five sub-sites have been separated within the one cutting.

The five sub-sites were all along the southern batter of the cutting to facilitate orientation of samples by sun-compass. There was good agreement between sun-compass and magnetic-compass bearings and hence the three samples for which sun-orientation was not possible were considered to have insignificant declination errors. Thirty-four oriented samples were taken using a portable field drill, and from these cores, 52 specimens were prepared.

Table II gives an indication of the distribution of specimens between sub-sites and between rock types. Thin-sections and polished-sections were prepared from offcut discs from the cores and from separately collected hand samples. Table II lists the results of NRM (natural remanent magnetization) measurements made on a PAR-SM2 spinner magnetometer. It may readily be seen that the overall mean is close to the present field direction. This, together with the similarity of directions between groups, suggests that some secondary magnetization may be present, and partial alternating field demagnetization was carried out.

Partial alternating field demagnetization

In an attempt to isolate a stable primary remanence (in this case a thermoremanence, TRM) partial alternating field (AF) demagnetization was carried out using a Schonstedt GSD/1 AF demagnetizer. A pilot specimen from each rock type at each sub-site was step-demagnetized at peak fields of 50, 100, 150, 200, 300, 400, 600 and 1000 oersteds. On the basis of both progressive change in direction and change in moment or intensity the most suitable AF peak field was selected for the treatment of the other specimens. Table II summarizes the results of the partial AF demagnetization, the directions being shown in Figure 2. No results have been discarded from the specimen directions plotted in Figure 2.

TABLE II

Palaeomagnetic results for the Yetholme adamellite and related rocks

II*a*. *NRM results*

Group	AF field oersted	*N*	*D* degrees, T	*I* degrees	α_{95} degrees	*k*	*R*	Note
Adamellite	0	34	010	—57	5.4	22.0	32.5	
Xenoliths	0	8	004	—63	13.0	19.0	7.63	
Aplite	0	3	003	—67	29.1	19.0	2.90	
Dolerite	0	7	003	—62	4.7	165.	6.96	
Sub-site 1	0	7	014	—57	12.1	25.9	6.77	
Sub-site 2	0	8	012	—47	15.3	14.0	7.50	
Sub-site 3	0	12	020	—61	9.2	23.1	11.5	
Sub-site 4	0	2	011	—60	—	—	1.96	1
Sub-site 5	0	23	356	—62	4.4	47.7	22.5	
All	0	52	008	—59	4.1	24.2	49.9	

II*b*. *Results of partial AF demagnetization*

Group	AF field oersted	*N*	*D* degrees, T	*I* degrees	α_{95} degrees	*k*	*R*	Note
Adamellite	150	34	356	—43	7.8	11.0	31.0	
Xenoliths	100 to 200	8	357	—57	14.4	15.7	7.55	
Aplite	200	3	016	—65	39.3	10.9	2.82	
Dolerite	200	7	354	—55	9.2	43.9	6.86	
Sub-site 1	—	7	012	—49	15.8	15.5	6.61	
Sub-site 2	—	8	001	—34	21.7	7.5	7.07	
Sub-site 3	—	12	006	—54	12.5	13.1	11.2	
Sub-site 3 (no aplite)	—	9	006	—50	14.4	13.7	8.41	
Sub-site 4	—	2	348	—43	—	—	1.97	1
Sub-site 5	—	23	347	—50	7.8	16.0	21.6	
All (no aplite)	—	49	356	—47	6.0	12.5	45.2	
All	—	52	357	—49	5.9	12.2	47.8	
Present geomagnetic field at Yetholme			011	—64				

Explanation: *N* is the number of specimens used in the Fisher (1953) analysis to determine the mean direction (declination *D*, inclination, *I*). α_{95} is the half-angle of the cone of 95% confidence, *k* the estimate of precision and *R* the length of the resultant vector (Fisherian statistics).

Note 1: Insufficient data. (All other means can be said to be statistically significant at the 95% level using Watson's (1956) criterion.)

The scatter of directions has increased slightly after partial AF demagnetization, but at the same time that the mean direction has moved away from the present geomagnetic field direction (Table II). This suggests that the NRM contained significant viscous (VRM) and isothermal (IRM) components which have been removed to reveal a TRM. It is possible that not all VRM and IRM components have been removed. Separation of the aplite results allowed further assessment of the stability of magnetization of the aplite. Although only three aplite specimens were measured, the scatter of directions and the closeness of the mean direction to that of the geomagnetic field suggest instability of magnetization. Because several rock types of similar, although slightly different, age were measured different points on the secular variation curve may have been sampled. If, as noted by McElhinny (1973, p. 29), 'secular variation is significantly smaller in pre-Tertiary times than in Tertiary', then the results for these Carboniferous rocks may have averaged out secular variation.

Using specimens as units (after partial AF demagnetization and excluding the three aplite specimens) (Table IIb), and the dipole hypothesis, a pole position was calculated to be at 59°S, 150°E ($dp = 5°$, $dm = 8°$).

DISCUSSION AND CONCLUSIONS

A combined palaeomagnetic, petrological, geochemical, and radiometric dating investigation of a small adamellite stock at Yetholme has provided evidence in support of a Late Carboniferous age for its emplacement. This is consistent with the idea that it is a satellitic intrusion of the Bathurst batholith.

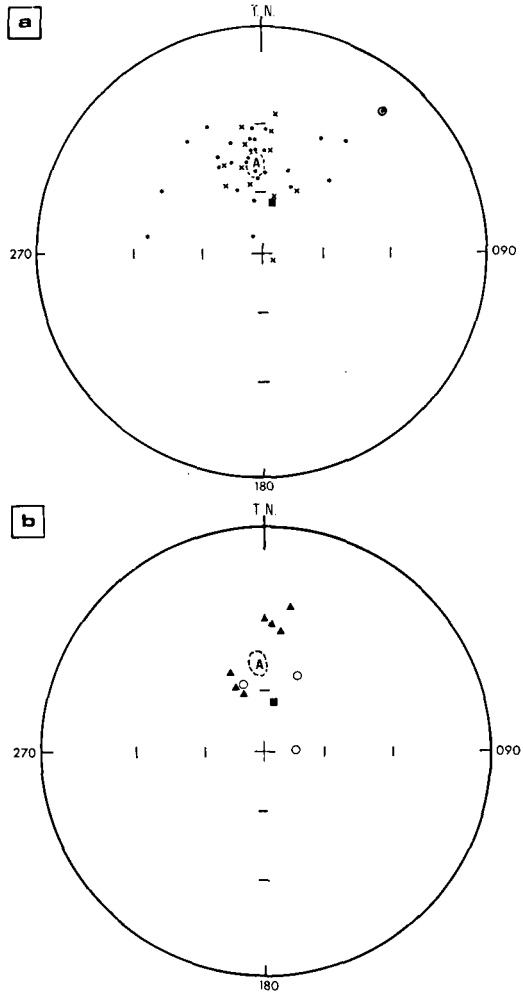


Fig. 2. Palaeomagnetic directions for all the specimens after partial AF demagnetization, plotted on the upper hemisphere of a stereographic equal-angle projection—that is, all but the circled point (lower hemisphere) are 'normal' directions. The direction of the geomagnetic field at the sampling locality is shown by a square, also plotted on the upper hemisphere. 'A' is the overall mean direction using specimens as units excluding the aplite (see Table IIb), the circle of 95% confidence being shown. (a) Adamellite (dot) and xenolith (cross) results. (b) Aplite (open circle and dolerite (triangle) results.

The Carboniferous age assigned to the Bathurst batholith and Yetholme adamellite is in agreement with radiometric ages. A Late Carboniferous age for the Yetholme adamellite is also supported by the palaeogeomagnetic pole position. Figure 3 shows the present result plotted on the apparent polar-wander path of Embleton *et al.* (1974, fig. 1) for the late Palaeozoic for southeastern Australia. The pole position for the Yetholme adamellite is very close to the pole positions for the Upper Carboniferous rocks studied by Irving (1966), whose results were based on studies of sedimentary and volcanic rocks sampled in the Newcastle to Tamworth region of New South Wales. Emplacement of the Bathurst batholith could have occurred as a result of what Irving (1966) termed the *'mid-Carboniferous continental drift episode'*.

Apparently the xenolith material has been remagnetized when incorporated into the adamellite and it possesses a TRM (or a CRM parallel to the TRM in the host adamellite). Granitic intrusions may cool sufficiently slowly to average out secular variation, but the grain-size of the iron-(titanium-) oixdes is generally large. These large grains are multidomain and are characterized by a low coercivity spectrum. Although the results of this investigation are based on studies of four rock types, the adamellite constitutes the largest proportion of the sample population. The Yetholme adamellite apparently possesses sufficient fine-grained magnetite, some in aggregates, to retain a proportion of the original magnetization during partial AF demagnetization. The adamellite is

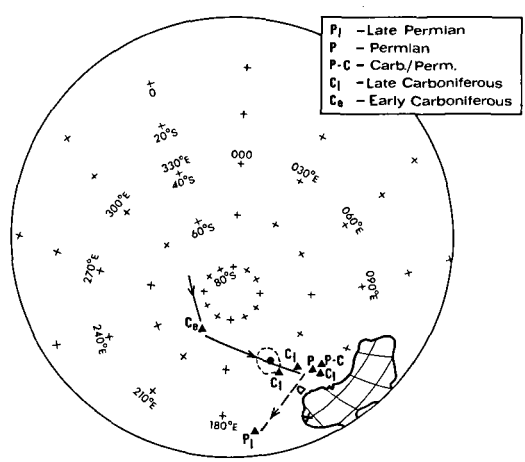


Fig. 3. Paleogeomagnetic pole position for the Yetholme adamellite (dot) plotted on an apparent polar-wander path for southeastern Australia for the late Palaeozoic (adapted from Embleton *et al.*, 1974, fig. 1). Yetholme is indicated on the map of Australia by a cross.

cut by dolerite dykes, one of which also possesses a direction of magnetization close to that of the host adamellite (although there may be two generations of dolerite). An aplite dyke cutting the adamellite apparently possesses an unstable magnetization.

When results are combined from all 49 specimens prepared from the three rock types of the Yetholme adamellite which possess stable magnetization, a mean direction of magnetization of $D = 356°$T, $I = 47°$ ($a_{95} = 6.0°$) is obtained. The Late Carboniferous (312 m.y.) palaeogeomagnetic pole corresponding to this magnetization was situated at 59°S, 150°E ($dp = 5°$, $dm = 8°$).

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REFERENCES

AYRES, D., 1974: Distribution and occurrence of some naturally-occurring polytypes of molybdenite in Australia and Papua New Guinea. *J. geol. Soc. Aust., 21,* pp. 273-278.

CAS, R. A. F., FLOOD, R. H., & SHAW, S. E., 1976: Hill End Trough: New radiometric ages: *Search, 7,* pp. 205-206.

EMBLETON, B. J. J., MCELHINNY, M. W., CRAWFORD, A. R., & LUCK, G. R., 1974: Palaeomagnetism and the tectonic evolution of the Tasman Orogenic Zone. *J. geol. Soc. Aust., 21,* pp. 187-193.

EVERNDEN, J. F., & RICHARDS, J. R., 1962: Potassium argon ages in eastern Australia. *J. geol. Soc. Aust., 9,* pp. 1-49.

FISHER, R., 1953: Dispersion on a sphere. *Proc. R. Soc., 217A,* pp. 295-305.

HIRT, B., HERR, W., & HOFFMEISTER, W., 1963: Age determinations by the rhenium-osmium method *in Radioactive Dating.* Internat. Atomic Energy Agency, Vienna, pp. 35-43.

IRVING, E., 1966: Paleomagnetism of some Carboniferous rocks from New South Wales and its relation to geological events. *J. geophys. Res., 71,* pp. 6025-6051.

MACKAY, R. M., 1964: The Yetholme granite. A study in the genesis of potash feldspar megacrysts. *Ph.D. thesis, Univ. Sydney* [unpubl.].

MCELHINNY, M. W., 1973: *Palaeomagnetism and Plate Tectonics.* Cambridge Univ. Press, 358 pp.

MCELHINNY, M. W., & EMBLETON, B. J. J., 1974: Australian palaeomagnetism and the Phanerozoic plate tectonics of eastern Gondwanaland. *Tectonophysics, 22,* pp. 1-29.

PACKHAM, G. H., 1968: *Bathurst 1:250 000 geological series sheet SI 55-8.* Geol. Surv. N.S.W.

PHILLIPS, E. R., & CARR, G. R., 1973: Myrmekite associated with alkali feldspar megacrysts in felsic rocks from New South Wales. *Lithos, 6,* pp. 245-260.

ROBERTSON, W. A., & WATKINSON, D. H., 1974: Paleomagnetism of the Spanish River alkalic rock-carbonatite complex, Twps. 107 and 108, Ontario. *Can. J. Earth Sci., 11,* pp. 795-800.

VALLANCE, T. G., 1969: Southern and central highlands fold belt. Plutonic and metamorphic rocks (*in The Geology of New South Wales,* G. H. Packham ed.). *J. geol. Soc. Aust., 16,* pp. 180-200.

WATSON, G. S., 1956: A test for randomness of directions. *Mon. Not. R. astr. Soc. geophys. Suppl., 7,* pp. 160-161.

WATTS, B. J., 1969: Geology of the Mt Tennyson area, south of Yetholme, N.S.W. *Proc. Linn. Soc. N.S.W., 94,* pp. 81-99.

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